

```
TYPE DATE POSSREP{  
    C CHAR  
    CONSTRAINT LENGTH (C) <= 8  
    AND  
    IS_DIGITS(C)  
};
```

```
TYPE TIME POSSREP{  
    c CHAR  
    CONSTRAINT LENGTH(c) <= 5  
};
```

```
TYPE IDFOUR POSSREP{  
    c CHAR  
    CONSTRAINT LENGTH(c)<= 5  
};
```

```
TYPE PDL POSSREP{  
    c CHAR  
    CONSTRAINT LENGTH(c)<= 14  
};
```

```
TYPE CODEPOSTAL POSSREP{  
    c CHAR  
    CONSTRAINT LENGTH(c) <= 5  
};
```

VAR client REAL RELATION{

id_client int,

nom_cli char,

prenom_cli char,

adresse1_cli char,

adresse2_cli char,

adresse3_cli char,

code_postal_cli CODEPOSTAL,

ville_cli char,

telephone_cli char,

mail_cli char

} KEY {id_client};

VAR compteur REAL RELATION{

pdl_prm PDL,

adresse1 char,

adresse2 char,

adresse3 char,

code_postal CODEPOSTAL,

ville char,

date_hiver_ete char,

date_ete_hiver char

} KEY {pdl_prm};

VAR contrat REAL RELATION{

no_contrat char,

date_debut_contrat int

```
} KEY {no_contrat};
```

```
VAR formule_tarifaire REAL RELATION{
```

```
    id_formule                                int,
```

```
    lib_formule                                char,
```

```
    descriptif_formule                        char,
```

```
    date_commercialisation                    int
```

```
} KEY {id_formule};
```

```
VAR puissance REAL RELATION {
```

```
    val_puissance                            int
```

```
} KEY {val_puissance};
```

```
VAR periode_conso REAL RELATION {
```

```
    id_periode_conso    char,
```

```
    lib_periode_conso    char
```

```
}    KEY{id_periode_conso};
```

```
VAR creneau REAL RELATION {
```

```
    no_creneau_hebdo    int,
```

```
    jour_hebdo            char,
```

```
    heure_debut            char,
```

```
    heure_fin              char
```

```
} KEY {no_creneau_hebdo};
```

```
VAR fournisseur REAL RELATION {  
    id_fournisseur          IDFOUR,  
    libelle_fournisseur     char,  
    telephone_fournisseur   char,  
    mail_fournisseur        char  
}    KEY {id_fournisseur};
```

```
VAR Date REAL RELATION {  
    date    DATE  
}    KEY {date};
```

```
VAR creneau_quotidien REAL RELATION{  
    heure        TIME  
} KEY{heure};
```

```
VAR comprend REAL RELATION{  
    no_creneau_hebdo      int,  
    id_formule            int  
}KEY{no_creneau_hebdo,id_formule};
```

```
CONSTRAINT comprend_fk_creneau  
comprend{no_creneau_hebdo} <= creneau{no_creneau_hebdo};
```

CONSTRAINT comprend_fk_formule

comprend{id_formule} <= formule_tarifaire{id_formule};

VAR posseder REAL RELATION{

id_client int,

pdl_prm PDL

}

KEY{id_client,pdl_prm};

CONSTRAINT posseder_fk_client

posseder{id_client} <= client{id_client};

CONSTRAINT posseder_fk_compteur

posseder{pdl_prm} <= compteur{pdl_prm};

VAR client_contrat REAL RELATION{

id_client int,

no_contrat char

}

KEY{id_client,no_contrat};

CONSTRAINT client_contrat_fk_client

client_contrat{id_client} <= client{id_client};

CONSTRAINT client_contrat_fk_contrat

client_contrat{no_contrat} = contrat{no_contrat};

VAR compteur_contrat REAL RELATION{

 pdl_prm PDL,

 no_contrat char

}

KEY{pdl_prm,no_contrat};

CONSTRAINT compteur_contrat_fk_compteur

compteur_contrat{pdl_prm} <= compteur{pdl_prm};

CONSTRAINT compteur_contrat_fk_contrat

compteur_contrat{no_contrat} = contrat{no_contrat};

```

VAR souscrire REAL RELATION{
    no_contrat char,
    date DATE,
    id_formule int
}
KEY{no_contrat,date,id_formule};

CONSTRAINT souscrire_fk_contrat
souscrire{no_contrat} <= contrat{no_contrat};

CONSTRAINT souscrire_fk_date
souscrire{date} <= Date{date};

CONSTRAINT souscrire_fk_formule_tarifaire
souscrire{id_formule} <= formule_tarifaire{id_formule};

```

```

VAR releve REAL RELATION {
    heure          TIME,
    pdl_prm        PDL,
    date           DATE,
    conso_kwh      int
} KEY{heure, pdl_prm, date };

```

```

CONSTRAINT releve_fk_date
releve{date} <= Date{date};

```

```
CONSTRAINT releve_fk_compteur
releve{pdl_prm} <= compteur{pdl_prm};
```

```
CONSTRAINT releve_fk_creneau_quotidien
releve{heure} <= creneau_quotidien{heure};
```

```
VAR tarifAbonnement REAL RELATION {
    val_puissance int,
    id_formule      int,
    prix_abo_ht      int,
    prix_abo_ttc int
} KEY { val_puissance, id_formule};
```

```
CONSTRAINT tarifAbonnement_fk_formule_tarifaire
{id_formule} <= formule_tarifaire{id_formule};
```

```
CONSTRAINT tarif_abonnement_fk_puissance
{val_puissance} <= puissance{val_puissance};
```

```
VAR periode_conso_creneau REAL RELATION{
    id_periode_conso char,
    no_creneau_hebdo int
} KEY{id_periode_conso,no_creneau_hebdo};
```


CONSTRAINT periode_conso_creneau_fk_creneau

periode_conso_creneau{no_creneau_hebdo} <= creneau{no_creneau_hebdo};

CONSTRAINT periode_conso_creneau_fk_periode_conso

periode_conso_creneau{id_periode_conso} = periode_conso{id_periode_conso};

VAR tarif_kWh REAL RELATION {

 prix_abo_ht int,

 prix_abo_ttc int,

 id_formule int,

 val_puissance int,

 id_periode_conso char

} KEY{id_formule,val_puissance,id_periode_conso};

CONSTRAINT tarif_kWh_fk_puissance

tarif_kWh{val_puissance} <= puissance{val_puissance};

CONSTRAINT tarif_kWh_fk_periode_conso

tarif_kWh{id_periode_conso} <= periode_conso{id_periode_conso};

CONSTRAINT tarif_kWh_fk_formule_tarifaire

tarif_kWh{id_formule} <= formule_tarifaire{id_formule};

```
VAR inclure REAL RELATION{
```

```
    id_formule int,
```

```
    id_periode_conso char
```

```
}
```

```
KEY{id_formule,id_periode_conso};
```

```
CONSTRAINT inclure_fk_formule_tarifaire
```

```
inclure{id_formule} <= formule_tarifaire{id_formule};
```

```
CONSTRAINT inclure_fk_periode_conso
```

```
inclure{id_periode_conso} <= periode_conso{id_periode_conso};
```

```
VAR proposer REAL RELATION{
```

```
    id_fournisseur IDFOUR,
```

```
    id_formule int
```

```
}
```

```
KEY{id_fournisseur,id_formule};
```

```
CONSTRAINT proposer_fk_formule
```

```
proposer{id_formule} <= formule_tarifaire{id_formule};
```

```
CONSTRAINT proposer_fk_fournisseur
```

```
proposer{id_fournisseur} <= fournisseur{id_fournisseur};
```